

Hazem Hassan

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PROFESSIONAL SUMMARY

Senior ML Engineer / AI Technical Lead with 9+ years of experience building production computer vision, OCR, video analytics, LLM/RAG, and data-centric AI pipelines. Strong hands-on experience developing semi-automated labeling workflows, teacher-model pseudo-labeling, uncertainty-driven review loops, dataset quality validation, and scalable ML data preparation infrastructure. Experienced in deploying end-to-end ML systems across cloud and edge environments using Python, PyTorch, TensorFlow, FastAPI, Docker, Kubernetes, ONNX/TensorRT, GCP, and AWS.

ROLES AND SPECIALITIES

Roles: Senior ML Engineer, AI Technical Lead, Senior Computer Vision Engineer.

Specialties: Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, Noisy Label Detection, Teacher-Student Pseudo-Labeling, Computer Vision Pipelines, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, ML APIs, MLOps, Cloud/Edge Deployment.

SKILLS

Data Engineering: SQL, PostgreSQL, NoSQL Basics, Pandas, NumPy, Data Cleaning, Data Transformation, Large-Scale Image/Video/Text Data Processing, Dataset Preparation Pipelines, Feature Engineering.

Frameworks: PyTorch, TensorFlow, Scikit-learn.

ML/DL: Model Training, Model Fine-Tuning, Feature Engineering, Hyperparameter Tuning, Custom Loss Function Design, Evaluation Metrics, Error Analysis.

Computer Vision & NLP: Object Detection, Segmentation, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, Anomaly Detection, Face Recognition, Liveness Detection, Image Processing, Optical Flow.

GenAI & LLMs: Large Language Models (LLMs), Multi-RAG, Retrieval-Augmented Generation, Embeddings, Reranking, Chunking Strategies, Prompt Engineering, LLM Evaluation.

Diffusion Models: Image Generation, Image Editing, Controllable Image Synthesis.

Deployment & MLOps: Model Serving, FastAPI, REST APIs, Docker, Kubernetes, CI/CD, Model Optimization, ONNX, TensorRT, Inference Performance Tuning.

Cloud & Tools: GCP, AWS, Git, Linux, Vertex AI, SageMaker/Ground Truth, Jupyter.

Edge AI: NVIDIA Jetson, Real-time Inference, Low-latency Video Pipelines.

Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI

March 2024—Present, Remote, EMEA

- Deployed real-time video analytics on NVIDIA Jetson edge devices across 10+ camera feeds / 5+ sites, sustaining 25–35 FPS with <150 ms latency for tracking, counting, and behavior analytics.
- Developed thermal/IR image-processing algorithms: fog dehazing/defogging to restore visibility in low-visibility conditions, and optical-flow-based detection of anomalous motion on IR camera feeds.
- Delivered embedded analytics for Single/Double Flow “Easier Gates”, achieving 94% flow accuracy and reducing ID switches by 14% through optimized detection + MOT tuning.
- Built PPE compliance detection (Aramco), achieving 94% precision / 90% recall, reducing false alarms by 25% via post-processing and threshold calibration.
- Built CCTV violence detection (spatio-temporal modeling) reaching F1 ~84% from only 10 minutes of labeled data, cutting annotation time by ~80% via a semi-automated labeling workflow.
- Designed active-learning style data selection loops for video analytics use cases, prioritizing high-uncertainty, edge-case, and failure-scenario samples for human review instead of full manual annotation.

Technical Lead

IDefy (Digital Identity/eKYC)

October 2022—February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI

January 2021—October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI

January 2017—January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

Co-Founder

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP

Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay

Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50-72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University – Electrical Electronics and Communication Engineering • 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition

Conference Paper, December 2019 – IEEE ASRU 2019

Prognostic AI for Post-Operative Infections

Conference Paper, November 2023 – FCI Annual Conference

QuranScript Verification System

Conference Paper, November 2016 – Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming

Gaming

Padel